



Society of Petroleum Engineers

SPE-227569-MS

Case History of Applying Business Model Canvas in the Complex Integrated Unconventional Project in the Middle East

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

The Lump Sum Turnkey (LSTK) Unconventional Project demands a robust project management team. Utilizing the Business Model Canvas (BMC) to coordinate and align various product lines has demonstrated effectiveness in addressing numerous challenges, particularly those related to the supply chain, throughout the project life cycle.

In this discussion, we will examine the components of the Business Model Canvas (BMC) which serve to illustrate how the project management team creates, delivers and captures value. The value proposition of the LSTK Unconventional Project Team is centered on delivering the utmost efficiency and exemplary job execution, with an unwavering commitment to safety for the client. This objective is realized predominantly through the pivotal roles of key resources, key activities, but above all, through strategic partnerships. To sustain the project's longevity, it is imperative to nurture customer relationships, particularly by soliciting feedback regarding service quality and committing to continuous improvement.

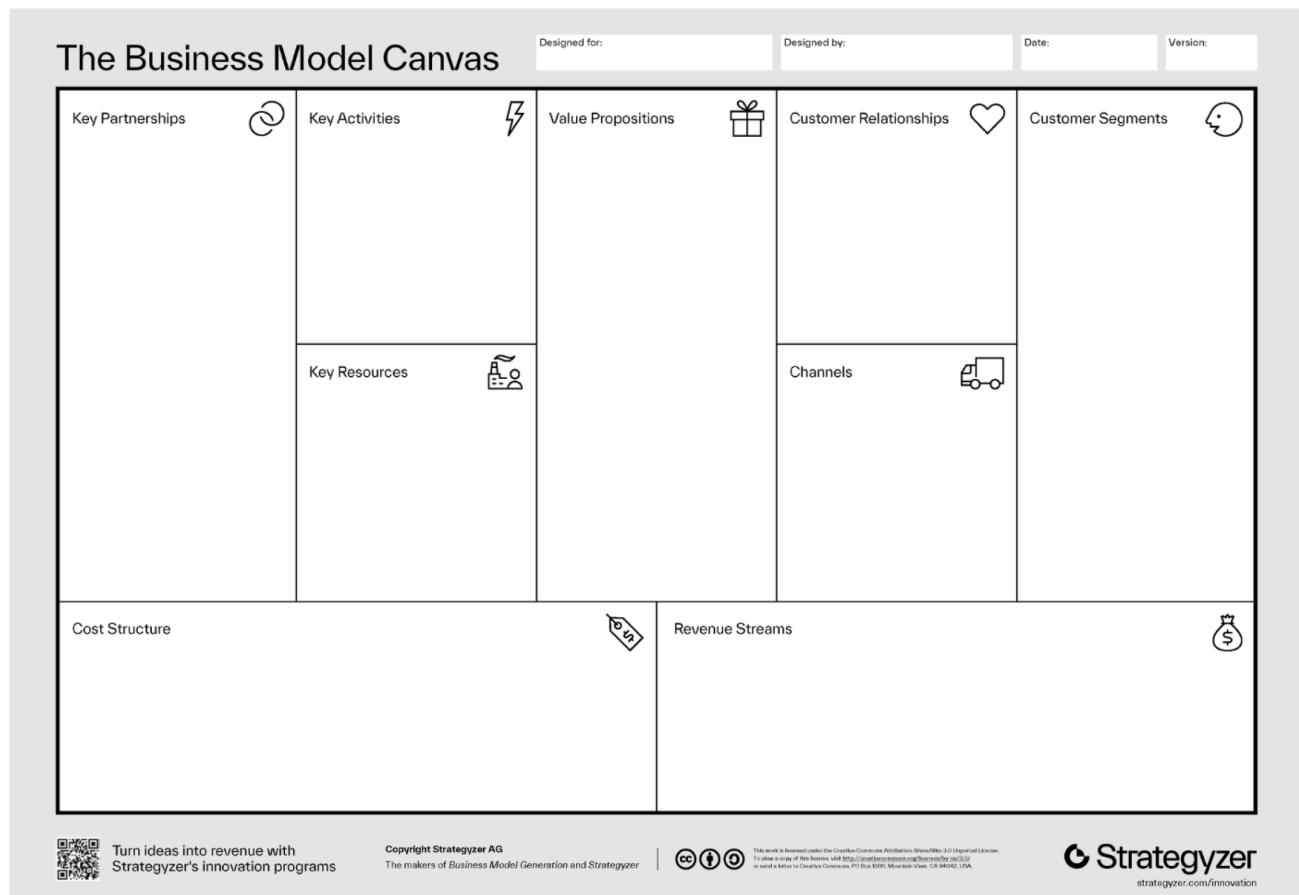
Business Model Canvas (BMC) proved beneficial at the outset of the project. Ongoing evaluation is crucial to ensure the business model aligns with the value proposition. Notable achievements include a positive trend in the number of jobs completed daily. Client involvement is essential for sustaining the deliverability of the value proposition, with service quality reviews conducted on a quarterly basis. To preserve a favorable margin between revenue and cost structure, adjustments and assessments of key partnerships, key activities, and key resources have been implemented. These changes are aimed at ensuring business continuity while upholding commitment to the customer's value proposition.

The Business Model Canvas (BMC) provides a comprehensive and exemplary framework for the organization, ensuring a consistent focus on the process involved in creating, developing, and delivering value to customers. It acts as a strategic tool that not only outlines but also reinforces the commitment to value generation and customer satisfaction within the business ecosystem.

Introduction

The Business Model Canvas (BMC), developed by Alexander Osterwalder and Yves Pigneur, is a strategic management and visual planning tool that provides a holistic framework for designing, analyzing, and refining business models. Comprising nine key building blocks—customer segments, value propositions,

channels, customer relationships, revenue streams, key resources, key activities, key partnerships, and cost structure—the BMC enables organizations to map out their operations and strategies in a clear, concise manner. Its versatility makes it applicable across industries, offering a structured yet adaptable approach to innovation and decision-making. In the oil and gas sector, where market dynamics, technological advancements, and environmental considerations create unique challenges, the BMC serves as an essential tool for aligning business strategies with operational realities and long-term goals.



Source: <https://www.strategyzer.com/library/the-business-model-canvas>

The Business Model Canvas (BMC) serves as a powerful strategic tool for companies in the oil and gas industry, enabling them to navigate the complexities of a volatile and capital-intensive market. By mapping out key components such as value propositions, customer segments, revenue streams, and cost structures, the BMC provides a clear framework for designing, evaluating, and refining business models. In the oil and gas sector, where exploration, production, and distribution involve intricate partnerships and high operational risks, the BMC helps firms align their resources and activities with market demands. For instance, upstream companies can use the canvas to optimize exploration strategies, while downstream players can focus on refining customer relationships and distribution channels, ensuring resilience and adaptability in a rapidly evolving energy landscape.

In the oil and gas industry, the Business Model Canvas (BMC) is increasingly utilized to foster innovation and address challenges such as fluctuating oil prices, regulatory pressures, and the transition to sustainable energy. By breaking down operations into nine key building blocks—ranging from key activities and partnerships to customer relationships and channels—the BMC enables companies to identify inefficiencies, explore new revenue opportunities, and pivot toward greener technologies. For example, midstream companies can leverage the canvas to streamline logistics and enhance supply chain partnerships,

while integrated firms can use it to balance traditional fossil fuel operations with investments in renewable energy. This structured yet flexible approach empowers oil and gas businesses to innovate and remain competitive in a dynamic global market.

Lump Sum Turnkey Project

Lump Sum Turnkey (LSTK) projects in the oil and gas industry represent a contractual arrangement where the contractor agrees to deliver a complete project—encompassing design, engineering, procurement, construction, and commissioning—for a fixed price and within a specified timeframe. This model is particularly prevalent in large-scale oil and gas developments, such as refineries, pipelines, and offshore platforms, where project certainty and cost control are paramount. By transferring significant risks, such as cost overruns and delays, to the contractor, LSTK contracts provide clients with predictability and reduced oversight responsibilities. However, the success of LSTK projects hinges on precise project scoping, robust risk management, and clear communication between stakeholders to avoid disputes and ensure alignment with technical and operational requirements.

The LSTK model offers several advantages for oil and gas companies, including cost certainty, streamlined project management, and reduced financial exposure to unforeseen expenses. By locking in a fixed price, operators can better forecast budgets and allocate resources efficiently, which is critical in a capital-intensive industry. Additionally, the turnkey nature of these contracts incentivizes contractors to optimize resources and adhere to schedules, as their profit margins depend on efficient execution. However, challenges arise when project specifications are ambiguous or when unforeseen geological or regulatory issues emerge, potentially leading to disputes or costly variations. Contractors, bearing the brunt of risks, must conduct thorough due diligence and maintain contingency plans, while clients must ensure comprehensive front-end engineering design (FEED) to minimize scope changes during execution.

LSTK contracts are strategically applied in oil and gas projects where speed, efficiency, and standardized processes are critical, such as in the development of modular facilities or brownfield expansions. For instance, in offshore oil platforms or liquefied natural gas (LNG) plants, LSTK contracts enable operators to leverage contractors' expertise in delivering complex, integrated systems under tight deadlines. The model is also favored in regions with stable regulatory frameworks, as it reduces the likelihood of external disruptions impacting project timelines. To maximize value, oil and gas companies often pair LSTK contracts with performance-based incentives, ensuring quality and safety standards are met. By fostering collaboration and aligning incentives, LSTK projects can drive operational excellence and deliver long-term value in a competitive energy market.

The application of LSTK contracts in unconventional fracturing (frac) projects, such as those in shale gas or tight oil reservoirs, holds significant potential but requires careful adaptation to the unique challenges of these operations. Unconventional frac projects involve complex, repetitive processes like hydraulic fracturing and well completion, where precision and scalability are critical. An LSTK approach could streamline the development of multi-well pads by integrating drilling, fracturing, and infrastructure construction under a single fixed-price contract, reducing coordination costs and accelerating production timelines. However, the variability of subsurface conditions and evolving environmental regulations necessitate robust risk-sharing mechanisms and flexible contract terms to address potential uncertainties. By leveraging advanced data analytics and standardized frac designs, operators and contractors can enhance the feasibility of LSTK in unconventional projects, driving cost efficiencies and enabling faster returns on investment in this rapidly growing sector.

Lump Sum Turnkey Project in Unconventional Frac

Opting for a Lump Sum Turnkey approach in an Unconventional Frac project is an effective solution that fosters a more integrated relationship between the client and the service company. The client benefits from

clear communication with the service company's integrated project team, which manages work distribution according to project needs

The unconventional Frac project can be divided into several scales of operations such as:

- **Pre-Intervention Work**
This scope involves preparing the location and well pad for fracturing operations. The critical success factor is ensuring that when the frac fleet arrives at the well pad, there is no downtime, allowing it to immediately begin the fracturing process.
- **Fracturing Work**
This scope involves the main operation. Fracturing plug and perf with Wireline pump down. This is a 24-hours operation non-stop creating fracturing stages in multi-well PAD horizontal wells. Plug-and-perf is a widely used technique in hydraulic fracturing for completing horizontal wells, particularly in unconventional oil and gas reservoirs like shale formations. It is a method to isolate and stimulate multiple sections (stages) of a well to maximize production.
- **Post-Intervention Work**
This scope encompasses Coil Tubing Milling and Well Flowback operations. Coil Tubing Milling involves removing all frac plugs used to isolate each fracturing stage. Following this, the well's productivity is evaluated through a well testing flowback operation, which helps estimate the production rate and determine the project's economic viability. In some cases, once the flowback is confirmed to be solids-free, the flowback line is linked to a centralized production facility at the Oil and Gas gathering station.

This paper will analyze each component of the business model canvas, providing a detailed explanation of every building block.

Building Block: Channels

This is the most important building block between service companies and oil companies to legally communicate and deliver products and services. In this channel building blocks, there are below key points:

1. Contract that defines detailed scope of work, duration, requirements,
2. Purchased Order that defines the amount of value customer requested on each specific scope of work
3. Quarterly performance meeting that defines the deliverables to the customer and reviewing challenges and gaps in executing the contract

Building Block: Customer Segments

In Business Model Canvas, the Customer Segments building block defines the different groups of people or organizations a business aims to reach and serve with its Value Proposition. It identifies who the customers are, what their needs are, and how they can be grouped based on common characteristics. This block is foundational because a business must clearly understand its target audience to tailor its offerings, channels, and relationships effectively.

For this unconventional frac project, these are defined customer segments:

1. Engineering team
Detailing the job program, technical requirements of each scope of work.
2. Operation team
Executing the job program as defined by engineering team

3. Planning team
Putting plan in place for all the pipeline of wells or pad available to be executed in the most efficient way
4. Support team
Supporting all the required things like water support, road access to the location, government permit for explosives and radioactive.
5. Compliance team
Ensuring all the executions are in compliance with safety and environmental regulations.
6. Contract team
Reviewing the invoice submitted by service providers that the job executed as per the requirements.

Building Block: Customer Relationships

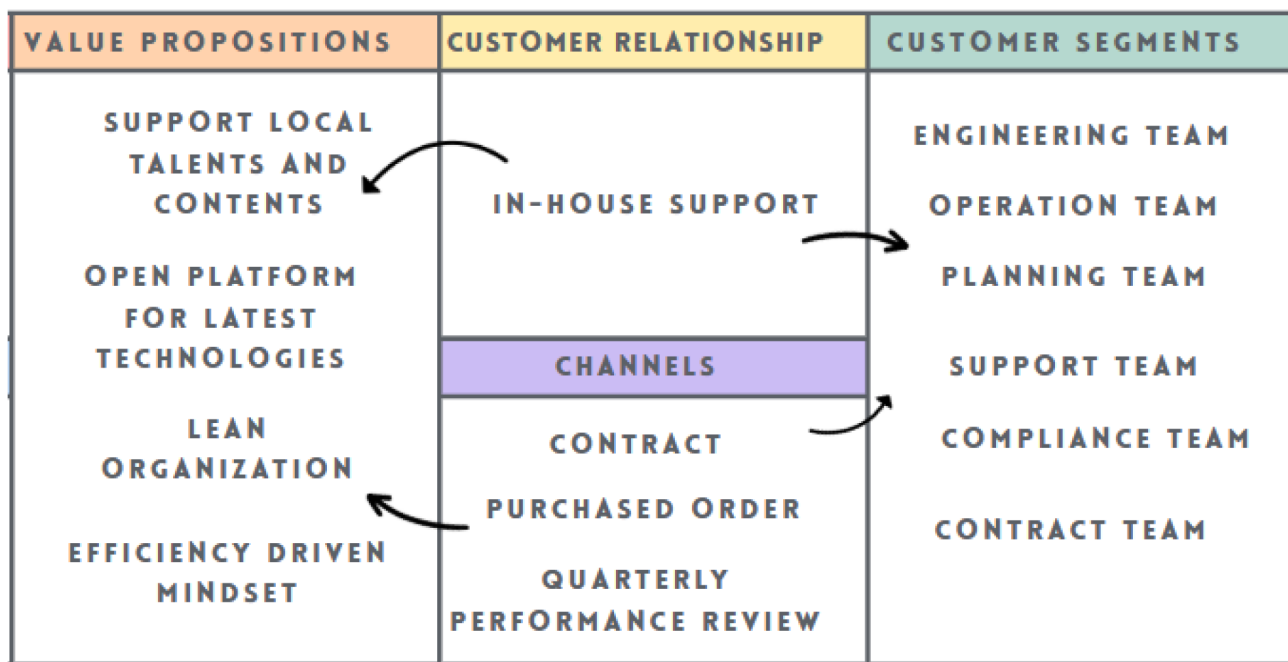
In this block, the dedicated in-house supports are provided that communicate intensely with customers in the Customer Segments building block. By having the in-house support, the information can be taken into account immediately and having quick respond action.

Building Block: Value Propositions

Value Propositions is the most critical building block in the Business Model Canvas since it will define the uniqueness of reason to keep customers relying on the service providers. Integrated Unconventional Project requires below value proposition to strengthen the position from the competition:

1. Support local talents and contents
The personnel operating the project will be majority of locals. Transitioning the product to be manufactured locally.
2. Open platform for latest technologies
Technologies enable us to make the project efficient. A service provider needs to facilitate the latest technology available and reliability that can easily be brought and adapted to the project.
3. Lean organization
Lean organization will minimize the time required to make changes, decision and technology adoption.
4. Efficiency driven mindset
Efficiency is key in the successful implementation of Integrated Unconventional Project. All the enablers, technologies implemented must bring efficiencies values to the equation.

Overall, 4 building blocks in delivering values to the customer and how to maintain the relationship between service providers and customers are visualized below:



Business Model Canvas will guide in the next building blocks on how to deliver the unique value propositions through several key building blocks:

1. Key Activities
2. Key Resources
3. Key Partnerships

Building Block: Key Activities

This paper utilizes the Business Model Canvas to define the Integrated Unconventional Project, organizing it into three distinct phases: Pre-Intervention Work, Fracturing Work, and Post-Intervention Work. These phases are strategically designed to ensure a cohesive approach to delivering value, with each scope of work seamlessly connecting to the Key Resources building block. Key Resources include the company's owned assets, such as frac fleets, wireline fleets, and coil-tubing fleets, as well as the specialized talent required to operate them. This integration allows the project to maintain operational control and efficiency, ensuring that each phase aligns with the project's objectives and meets the needs of the targeted Customer Segments. The project's success is further supported by the Key Partners building block, which outlines strategic collaborations critically to enhance the project's capabilities. These partners, including equipment suppliers, technology providers, and service contractors, complement the company's resources by providing specialized expertise or tools that optimize processes like fracturing or post-intervention evaluations. By linking the defined work scopes with both Key Resources and Key Partners, the project ensures a robust framework for delivering its Value Proposition, which focuses on high-performance, cost-effective, and tailored solutions. This structured approach positions the Integrated Unconventional Project to achieve operational excellence and strengthen client relationships through reliable and innovative service delivery.

Building Block: Key Resources

The Integrated Unconventional Project is designed to leverage a wide array of the company's existing assets to ensure efficient and effective execution. Central to this approach is the utilization of key resources, including company-owned service fleets such as frac fleets, wireline fleets, and coil-tubing fleets. These

assets provide the project with significant operational control, enabling streamlined processes across the defined work scopes of Pre-Intervention Work, Fracturing Work, and Post-Intervention Work. Additionally, the project relies on the specialized talent within the organization, whose expertise in operating these fleets ensures high-performance outcomes and alignment with the project's value proposition of delivering reliable and tailored unconventional energy solutions.

Beyond physical assets, the project's success is bolstered by the strategic integration of human resources and operational capabilities. The talent pool, skilled in managing and optimizing the company's fleets, plays a critical role in maintaining cost-effectiveness and operational excellence. By primarily utilizing company-owned services, the project minimizes dependency on external providers, enhancing flexibility and reducing costs. This resource-driven approach not only strengthens the project's ability to meet the needs of its customer segments but also supports the development of robust channels for client engagement, ensuring that the value proposition—centered on efficiency and customization—is consistently delivered throughout all phases of the project.

Building Block: Key Partnerships

Key Partnerships involve strategic alliances with external stakeholders, including equipment suppliers, technology providers, and specialized service contractors, who bring cutting-edge tools, innovative solutions, and niche expertise to enhance project outcomes

The overall building block of Key Activities, Key Resources and Key Partnerships are visualized below:

KEY PARTNERSHIP	KEY ACTIVITIES
SITE AND CAMP FRAC TREES AI AND DATA SCIENCE SAND DELIVERY WATER SUPPLY GAS DETECTION SYSTEM JOURNEY MANAGEMENT SYSTEM	PRE-INTERVENTION FRACTURING WORK POST-INTERVENTION
	KEY RESOURCES
	FRAC WIRELINE
	COIL WELL
	TUBING TESTING SLICKLINE

Building Block: Cost Structure

In Business Model Canvas, the Cost Structure building block outlines all the expenses a business incurs to operate its business model and deliver its Value Proposition to Customer Segments. It captures the financial implications of executing Key Activities, utilizing Key Resources, maintaining Key Partnerships, and engaging through Channels. For the Integrated Unconventional Project, which involves Pre-Intervention Work, Fracturing Work, and Post-Intervention Work in unconventional energy extraction, the cost structure

is critical to ensuring profitability while delivering high-performance, efficient, and customized solutions. Below is a detailed explanation of the Cost Structure building block, tailored to the project's context and aligned with the broader Business Model Canvas framework.

The purpose of the Cost Structure is that the Cost Structure defines the monetary costs required to sustain the business model, answering key questions such as:

- What are the most significant costs in operating the business?
- Which Key Resources and Key Activities drive the highest expenses?
- Are costs aligned with the Value Proposition (e.g., performance, customization, efficiency)?
- Is the business model cost-driven (focused on minimizing costs) or value-driven (focused on delivering premium value)?

For the Integrated Unconventional Project, the cost structure is primarily value-driven, emphasizing high-quality service delivery through company-owned assets (e.g., frac fleets, wireline fleets, coil-tubing fleets) and skilled talent. However, cost optimization remains essential to maintain competitiveness in the unconventional energy sector, balancing investments in performance with financial discipline.

The Cost Structure is interconnected with other elements of the Business Model Canvas, as costs stem from operating the business model:

- Key Activities: Planning, fracturing, and post-intervention analysis drive variable costs (e.g., fuel, materials) and fixed costs (e.g., salaries, software).
- Key Resources: Company-owned fleets and talent are major cost drivers, with maintenance, depreciation, and salaries forming significant fixed costs.
- Key Partnerships: Suppliers and contractors introduce variable costs for materials and services, reducing the need for in-house investment in niche areas.
- Channels: Direct client engagement and digital platforms incur marketing and IT costs to support transparent communication.
- Value Proposition: The focus on efficiency, performance, and customization justifies costs for advanced equipment, skilled labor, and technology.
- Customer Segments: Serving energy companies with complex needs requires investment in reliable assets and compliance, driving operational and regulatory costs.
- Revenue Streams: Costs must align with revenue from project contracts to ensure profitability, supporting premium pricing for high-quality services.

A well-defined Cost Structure ensures the Integrated Unconventional Project remains financially viable while delivering its Value Proposition. Poor cost management can erode profitability, limit scalability, or compromise service quality, weakening the project's competitive position. By aligning costs with Key Activities, Key Resources, and Key Partnerships, the project balances its value-driven approach with financial discipline, supporting sustainable growth and client satisfaction in the unconventional energy market.

For the Integrated Unconventional Project, the Cost Structure reflects a strategic focus on leveraging company-controlled resources and partnerships to deliver high-value services. This approach supports the Value Proposition of efficiency, performance, and customization, ensuring the project meets client expectations while maintaining economic sustainability.

Building Block: Revenue Stream

In the Business Model Canvas, the Revenue Streams building block represents the ways a business generates income from its Customer Segments by delivering its Value Proposition. It outlines the cash flows resulting

from the successful execution of Key Activities, utilization of Key Resources, and collaboration with Key Partnerships through effective Channels. For the Integrated Unconventional Project, which involves Pre-Intervention Work, Fracturing Work, and Post-Intervention Work in unconventional energy extraction, the Revenue Streams are critical to ensuring financial sustainability while delivering high-performance, efficient, and customized solutions. Below is a detailed explanation of the Revenue Streams building block, tailored to the project's context and aligned with the broader Business Model Canvas framework.

The Revenue Streams building block defines how the business monetizes its offerings, answering key questions such as:

- How does the project generate revenue from its Customer Segments?
- What value are customers willing to pay for, and how are they willing to pay?
- How do revenue streams align with the Value Proposition (e.g., efficiency, performance, customization)?
- Are revenue streams sustainable and sufficient to cover the Cost Structure?

For the Integrated Unconventional Project, the revenue streams are tied to providing specialized services in the unconventional energy sector, leveraging company-owned assets like frac fleets, wireline fleets, and coil-tubing fleets, as well as skilled talent. The project's Value Proposition—centered on operational efficiency, high-performance outcomes, and tailored solutions—drives the types of revenue streams, which are primarily value-driven, reflecting the premium quality and customization offered to energy companies.

A well-defined Revenue Streams building block ensures the Integrated Unconventional Project is financially viable, generating sufficient income to cover the Cost Structure while delivering its Value Proposition. Poorly designed revenue streams can lead to cash flow issues, reliance on a single client, or inability to sustain high-quality services, undermining the project's success. By aligning revenue streams with Customer Segments, Value Proposition, and operational capabilities, the project can achieve profitability, support growth, and build long-term client relationships.

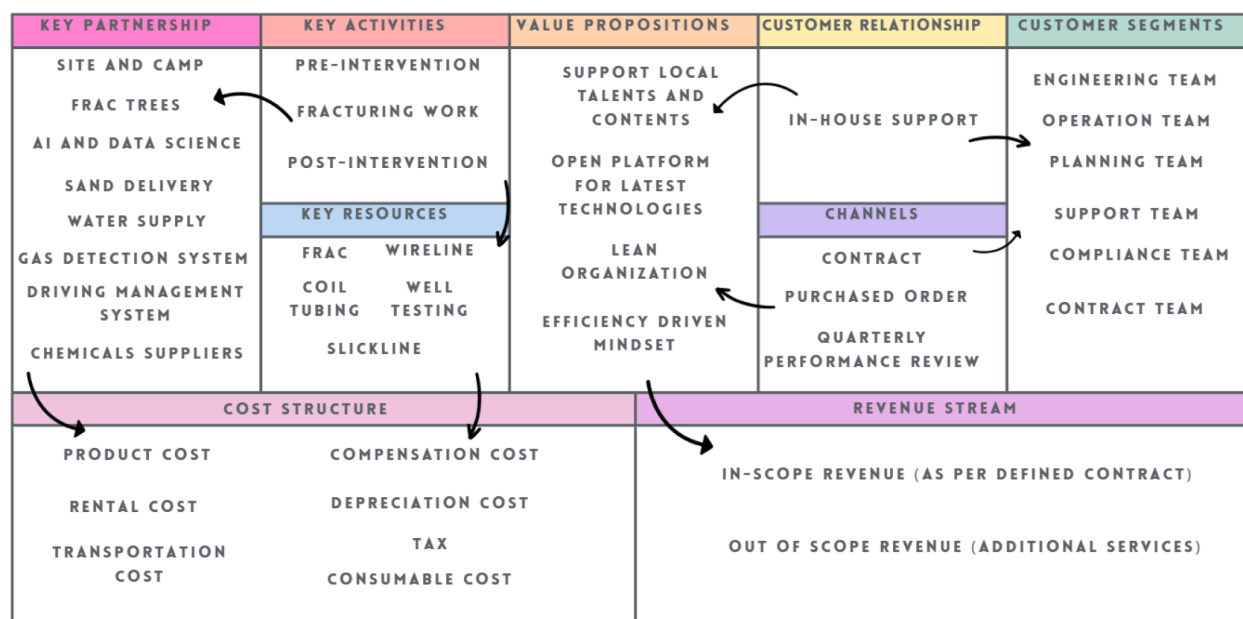
For the Integrated Unconventional Project, the Revenue Streams reflect a strategic focus on leveraging company-controlled resources and partnerships to deliver premium, customized services. This value-driven approach supports diverse income sources—service fees, contracts, consulting, and performance bonuses—ensuring financial resilience and alignment with the project's goal of operational excellence in the unconventional energy market.

The overall building block of Cost Structures and Revenue Streams:

COST STRUCTURE		REVENUE STREAM
PRODUCT COST	COMPENSATION COST	IN-SCOPE REVENUE (AS PER DEFINED CONTRACT)
RENTAL COST	DEPRECIATION COST	
TRANSPORTATION COST	TAX	
	CONSUMABLE COST	
		OUT OF SCOPE REVENUE (ADDITIONAL SERVICES)

The overall 9 building blocks of the Integrated Unconventional Project can be visualized below:

INTEGRATED UNCONVENTIONAL PROJECT – BUSINESS MODEL CANVAS



Summary

The Business Model Canvas provides an intuitive and visually accessible framework for mapping out the Integrated Unconventional Project, enabling stakeholders to easily grasp its structure and strategic approach. By organizing the project into nine key building blocks—Customer Segments, Value Proposition, Channels, Customer Relationships, Revenue Streams, Key Resources, Key Activities, Key Partnerships, and Cost Structure—the canvas presents a clear, one-page snapshot of how the project creates, delivers, and captures value. For instance, the project's focus on energy companies as Customer Segments, its Value Proposition of efficiency and customization, and the use of company-owned frac fleets as Key Resources are visually interlinked, making it simple to understand operational flow. This visual clarity is particularly valuable for aligning cross-functional teams, communicating with clients, and identifying gaps or opportunities in the project's strategy, ensuring all aspects, from Pre-Intervention to Post-Intervention Work, are cohesively represented.

The Business Model Canvas excels in capturing specific details tailored to the Integrated Unconventional Project, allowing for a granular breakdown of each building block to address the project's unique scope of work: Pre-Intervention Work, Fracturing Work, and Post-Intervention Work. For example, Key Activities include precise tasks like reservoir analysis, hydraulic fracturing, and post-intervention optimization, while Key Resources detail company-owned assets such as frac fleets, wireline fleets, and coil-tubing fleets, alongside specialized talent. The Revenue Streams specify income from service fees, project-based contracts, and performance bonuses, while the Cost Structure outlines fixed costs and variable costs. Key Partnerships with technology providers and suppliers are highlighted to enhance innovation. This specificity ensures that every operational and financial detail is accounted for, enabling precise planning and execution in the unconventional energy sector.

Beyond visualization and detail, the Business Model Canvas fosters strategic flexibility and stakeholder alignment for the Integrated Unconventional Project, making it a dynamic tool for ongoing refinement and decision-making. It allows the project team to adapt to market changes, such as fluctuating energy prices or regulatory shifts, by revisiting building blocks like Revenue Streams (e.g., introducing subscription fees for ongoing monitoring) or Key Partnerships (e.g., collaborating with new environmental tech providers). The canvas also facilitates collaboration by providing a shared language for discussing trade-

offs, such as balancing the Cost Structure with premium pricing to maintain the Value Proposition of high-performance services. By integrating all elements—from Channels like direct client engagement to Customer Relationships built on transparency—the canvas ensures the project remains financially viable, operationally efficient, and competitive, driving long-term success in the unconventional energy market.